



**Politecnico
di Torino**

Vehicle Braking System Analysis and Optimization

Bachelor's Degree in Automotive Engineering

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Abstract

The purpose of this report is to analyze the braking systems of a vehicle. The study begins by considering an ideal braking behavior, and then explores an improvement introduced by adding an additional component to the system. This enhancement is achieved by examining the different operating conditions of the front and rear brakes, and by implementing a relief valve to regulate the pressure distribution between them. In general, passenger car braking systems are designed with safety as a priority; therefore, the braking system is developed to ensure that, whenever the driver applies the brakes, the vehicle decelerates both effectively and safely.

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1 Introduction

The analysis of the vehicle braking system is divided into two parts: first, considering ideal braking conditions, and then evaluating performance under real operating conditions. Since the braking system plays a crucial role in vehicle safety, it is engineered with built-in redundancy to ensure reliability. The free-body diagram of the vehicle considered in this study is presented in Figure 1.

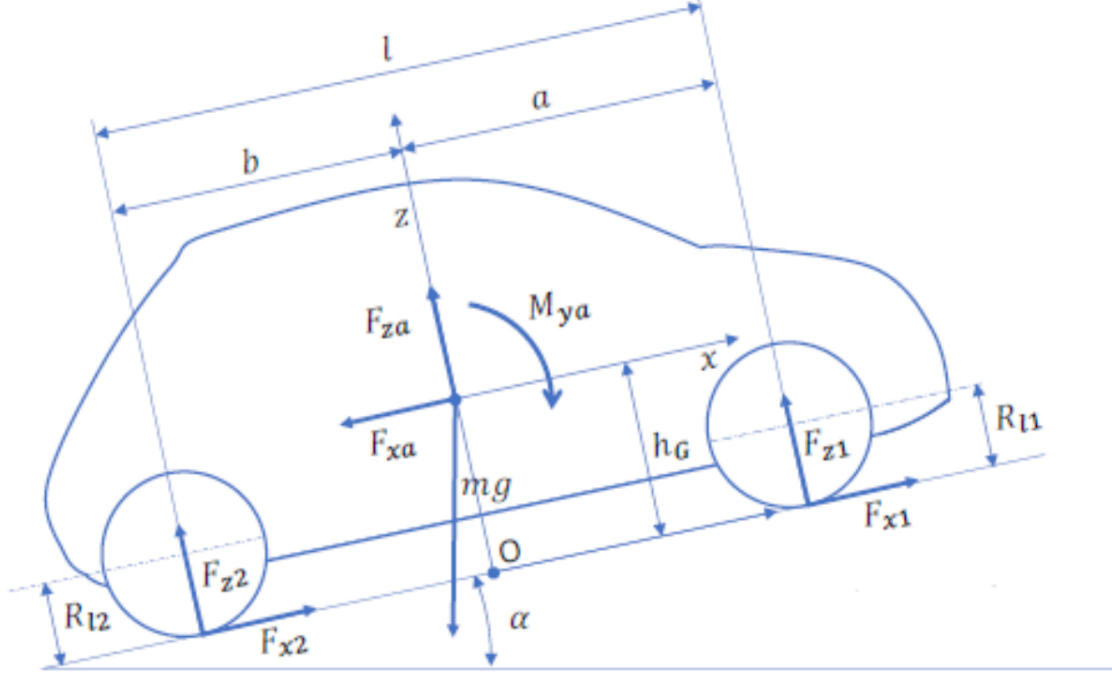


Figure 1: Vehicle's Free Body Diagram

For the analysis, two different vehicle mass configurations will be considered. The first one, referred to as mass B (m_B), includes the vehicle mass, the driver, and standard baggage. The second configuration, referred to as mass F (m_F), consists of the vehicle mass B plus four passengers and additional heavy baggage. The main parameters of the vehicle model are illustrated in Figure 2. The braking system analysis will be performed for each vehicle configuration.

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%% Parameters of vehicle

hG = 0.48; % height of the center of gravity
m = 1750; % mass vehicle [kg]
g = 9.81;
l = 2.72; % [m] wheel base
b = 1.4; % distance center of gravity - rear axle
alpha = 0; % slope
a = 1.32; % distance center of gravity - front axle

%% Mass added

md = 75; % driver mass
mb = 10; % baggage mass
mB = m + md + mb; % mass standard B
m4 = 280; % mass 4 passengers
mF = mB + m4 + 4*mb; % mass standard F
```

Figure 2: Main Vehicle's parameters

2 Ideal Braking

The total braking force in a vehicle can be expressed as follows:

$$F_x = \sum_i \mu_{xi} F_{zi} \quad (1)$$

where the sum includes all the braking wheels. The rolling resistance and the aerodynamic forces are negligible with respect to the braking forces. Applying second Newton's law in the x-direction to the FBD's, the vehicle model can be written as:

$$m \frac{dv}{dt} = \sum_i \mu_{xi} F_{zi} - mgsin\alpha \quad (2)$$

The ideal braking condition is defined when the coefficients μ_{xi} have the same value for all the wheels. Then, replacing Equation 1 into Equation 2, assuming the road grade equal to zero ($\alpha = 0^\circ$) and dividing over the mass we obtain:

$$\frac{dv}{dt} = \mu_x g = \frac{\sum_i F_{xi}}{m} = \frac{F_{x1} + F_{x2}}{m} \quad (3)$$

where F_{x1} , F_{x2} are the longitudinal forces acting on the front and rear axle respectively, depending on the vertical forces on the tires. As it was stated before, the rolling resistance and aerodynamic forces were neglected, then the longitudinal loads are based exclusively on F_{zi} , F_{z2} .

$$F_{x1} = \mu_{x1} F_{z1} = \mu_{x1} \frac{m}{l} \left[gbcos\alpha - gh_G sin\alpha - h_G \frac{dV}{dt} \right] \quad (4)$$

$$F_{x2} = \mu_{x2} F_{z2} = \mu_{x2} \frac{m}{l} \left[gacos\alpha - gh_G sin\alpha + h_G \frac{dV}{dt} \right] \quad (5)$$

By substituting μ_x from the last two equations and applying the longitudinal equilibrium condition, the ideal braking equation is obtained.

$$(F_{x1} + F_{x2})^2 + mgcos\alpha \left(F_{x1} \frac{a}{h_G} - F_{x2} \frac{b}{h_G} \right) = 0 \quad (6)$$

2.1 Ideal braking curve

To plot the ideal braking curve, Equation 6 is used. For each vehicle configuration (B or F), the corresponding mass is substituted into the equation to obtain the green curve shown in Figure 3. As stated, ideal braking is assumed when the friction coefficients are equal and constant on both axles. Accordingly, curves for different friction coefficient values are generated. The values of μ_{x1} and μ_{x2} , ranging from 0.2 to 1.2, are applied as inputs in the following equations and represented with black and blue lines, respectively, in Figure 3 and Figure 4.

$$F_{x2} = \frac{mgb}{h_G} cos\alpha - \left(\frac{l}{\mu_{x1} h_G} + 1 \right) F_{x1} \quad (7)$$

$$F_{x2} = mg \frac{\mu_{x2} a}{l - \mu_{x2} h_G} cos\alpha + \left(\frac{\mu_{x2} h_G}{l - \mu_{x2} h_G} \right) F_{x1} \quad (8)$$

If the deceleration $\frac{dV}{dt}$ is assumed to be constant, the relationship between F_{x1} and F_{x2} is expressed by Equation 9. Constant values of $\frac{dV}{dt}$, ranging from $-0.2g$ to $-1.2g$, are used as inputs and represented with dashed blue lines in Figure 3 and Figure 4.

$$F_{x2} = mgsin\alpha + m \frac{dV}{dt} - F_{x1} \quad (9)$$

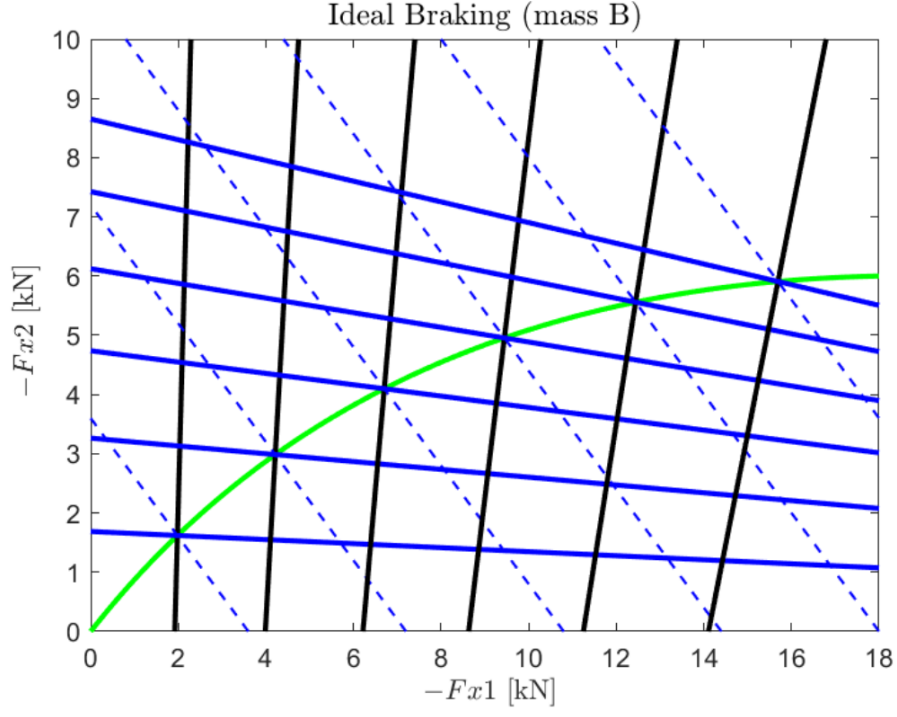


Figure 3: Ideal Braking Curve – mass B

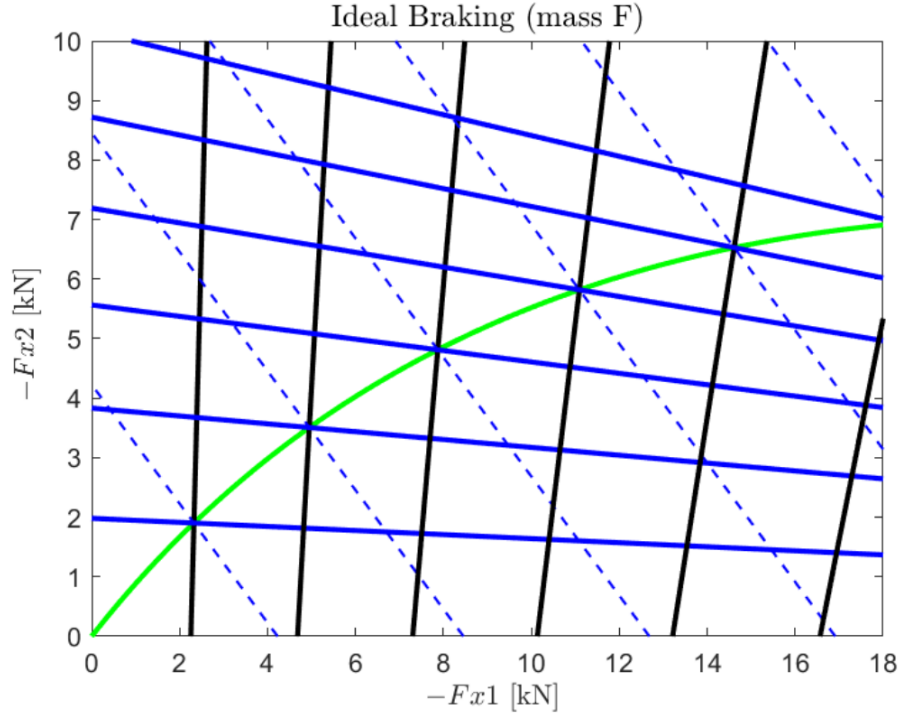


Figure 4: Ideal Braking Curve – mass F

3 Braking under actual conditions

In practice, the braking system does not perform in the same way as the ideal braking case. To represent the behavior of a real braking system, it is necessary to extend the analysis beyond the previous equations. Generally, the ratio between the braking torque on the front and rear axles is assumed to be constant and is described by the following equation:

$$K_b = \frac{M_{b1}}{M_{b2}} \quad (10)$$

where M_{b1} , M_{b2} are the braking torque in the front and rear axle respectively. If all the wheels have the same radius, the last equation could be further simplified as follows:

$$K_b = \frac{rF_{x1}}{rF_{x2}} = \frac{F_{x1}}{F_{x2}} \quad (11)$$

To better understand the vehicle's braking behavior, it is necessary to plot the actual braking curve, shown as the red line in Figure 5 and Figure 6, and compare it with the ideal braking curve. To begin designing the K_b coefficient, Equation 4 and Equation 5 can be substituted into Equation 11. In this project, the design point is defined as the condition where the braking efficiency is equal to 1 for both values in Equation 19, as explained in the dedicated section.

$$K_b = \frac{b - \mu_{x1}h_G}{l + \mu_{x1}h_G - b} = \frac{l - \mu_{x2}h_G - a}{\mu_{x2}h_G + a} \quad (12)$$

For this project, the operating point is chosen at $\mu_x = -0.4$. The corresponding K_b line is represented by the red line in Figure 5 and Figure 6, for the two vehicle masses, B and F.

Following the path of the actual braking system, as the braking force increases, the vehicle enters a zone of instability where oversteering may occur. For safety reasons, this behavior must be avoided. Therefore, modern braking systems are designed to modulate the braking forces, reducing the pressure on the rear axle. This is achieved through a valve that activates before the friction coefficients of the axles become equal, ideally at 90% of the intersection between the ideal braking curve and the linear braking curve.

With this modification, the braking system follows the purple path shown in Figure 5 and Figure 6. By implementing the proportional valve, the system becomes bilinear, operating within the safe zone, below the ideal braking curve. The braking valve line is designed from the triggering point A to point C; in other words, when the friction coefficient ranges from $-0.4 < \mu_x < -1$. The slope of the purple line can be defined as:

$$K_b' = \frac{\Delta F_{x1}}{\Delta F_{x2}} \quad (13)$$

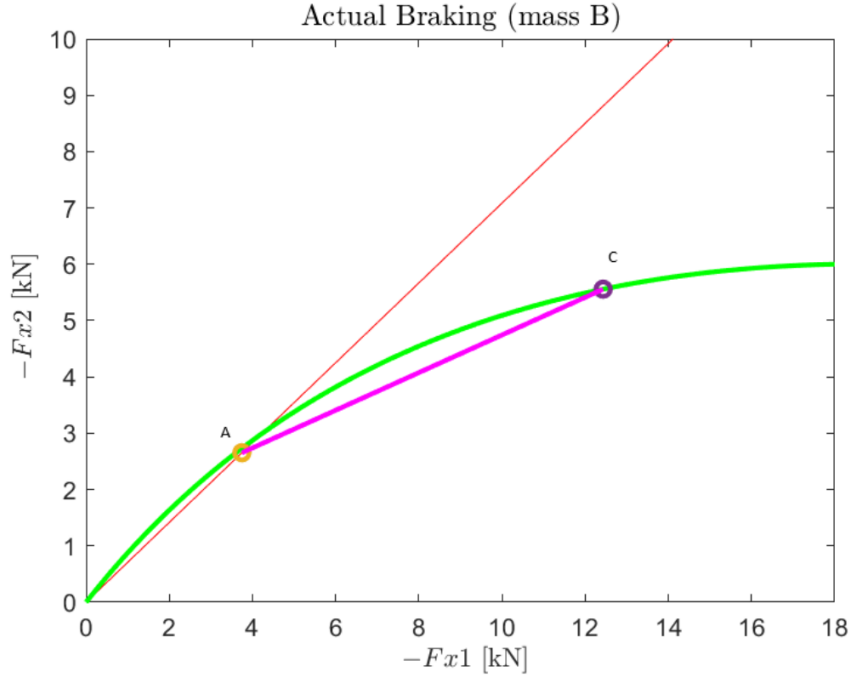


Figure 5: Actual braking system - mass B

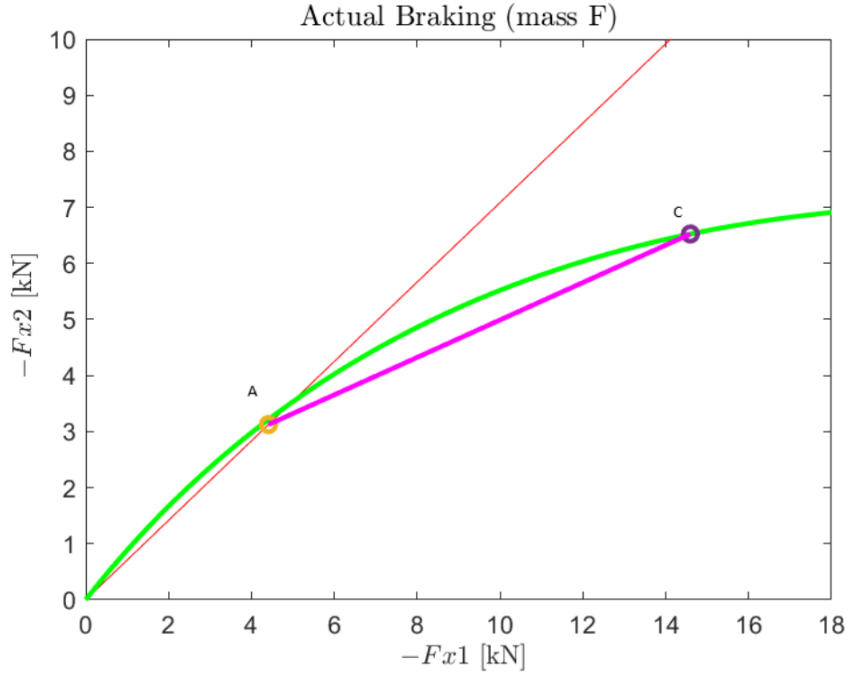


Figure 6: Actual braking system - mass F

The ratio coefficients for the vehicle of mass B and F are the following:

- $K_{bB} = 1.4113$
- $K_{bF} = 1.4113$
- $K'_{bB} = 2.9916$
- $K'_{bF} = 2.9945$

4 Braking System Efficiency

When designing the ratio of braking forces, represented by the K_b coefficient, safety is the primary consideration. Consequently, passenger car braking systems are designed to operate below the ideal braking curve. An efficiency term, η_b , can be defined to relate the actual deceleration to the ideal deceleration under ideal braking conditions.

$$\eta_b = \frac{\left(\frac{dV}{dt}\right)_{actual}}{\left(\frac{dV}{dt}\right)_{ideal}} = \frac{\left(\frac{dV}{dt}\right)_{actual}}{\mu_x g} \quad (14)$$

From Equation 3, the deceleration for the actual braking system can be determined. By combining it with Equation 14, the deceleration can be expressed in terms of F_{x1} and K_b .

$$\left(\frac{dV}{dt}\right)_{actual} = \frac{F_{x1} + F_{x2}}{m} = \frac{F_{x1} \left(1 + \frac{1}{K_b}\right)}{m} \quad (15)$$

The same procedure can be performed to obtain the deceleration in terms of F_{x2} and K_b .

$$\left(\frac{dV}{dt}\right)_{actual} = \frac{F_{x1} + F_{x2}}{m} = \frac{F_{x2} (1 + K_b)}{m} \quad (16)$$

Continuing replacing Equations 4 and 5 into Equations 15 and 16 respectively, it is obtained:

$$\left(\frac{dV}{dt}\right)_{actual} = \frac{\mu_{x1} g b (K_b + 1)}{l K_b + \mu_{x1} h_G (K_b + 1)} \quad (17)$$

$$\left(\frac{dV}{dt}\right)_{actual} = \frac{\mu_{x2} g a (K_b + 1)}{l - \mu_{x2} h_G (K_b + 1)} \quad (18)$$

Since the braking forces differ but remain proportional on both axles, the efficiency must be evaluated for each axle individually. The overall vehicle efficiency is then defined as the minimum efficiency between the two axles. By substituting the equations expressing the braking force on each axle into Equation 15, the following expression is obtained:

$$\eta_b = \min \left\{ \frac{b(K_b + 1)}{lK_b + \mu_{x1}h_G(K_b + 1)}, \frac{a(K_b + 1)}{l - \mu_{x2}h_G(K_b + 1)} \right\} \quad (19)$$

The efficiency of the braking system, withouth the partitioning valve, is computed for $\mu_x = -0.4$ and $\mu_x = -0.65$, and is represented for each vehicle mass, B and F, by the red and blue lines, respectively, in Figure 7 and Figure 8. Moreover, the curve showing the efficiency evolution achieved with the proportional valve is also depicted with the black line.

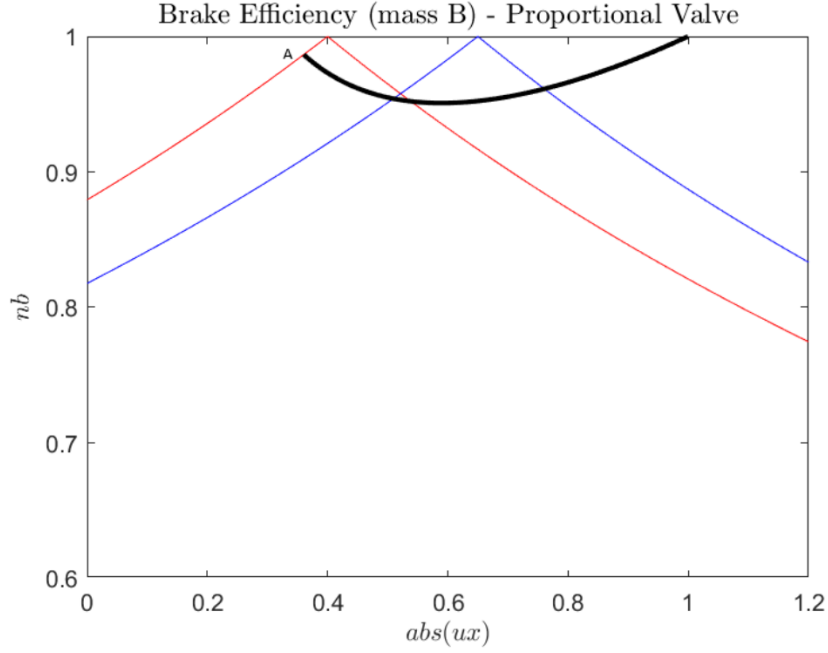


Figure 7: Brake efficiency - mass B

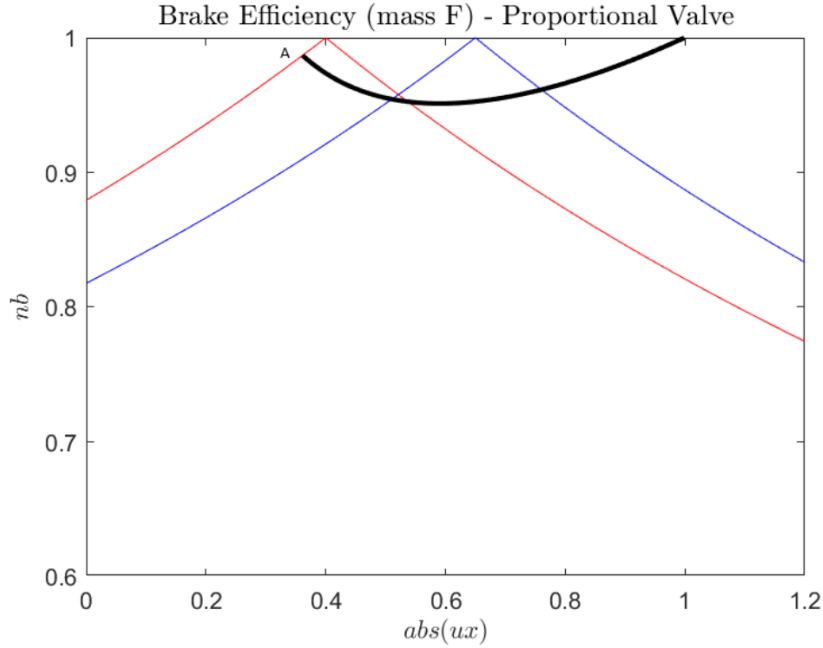


Figure 8: Brake efficiency - mass F

5 Conclusions

The implementation of a pressure-relief valve in the braking system helps achieve a better distribution of longitudinal forces during braking, thereby improving vehicle stability. This is reflected in the improved braking efficiency, shown by the black curve in Figure 7 and Figure 8. The minimum efficiency typically occurs at the front tires. Consequently, the valve releases pressure in the rear braking lines to prevent rear-wheel lockup, allowing the front tires to lock first while the rear tires retain some grip. This prevents the more dangerous scenario of rear-wheel lockup, which could lead to vehicle spin. Although this analysis does not capture every real-world transient effect, since the actuation of components is not instantaneous, it provides a conservative basis for designing a safe braking system.